An Actuated Arc–Leg Hexapod for Walking, Rolling, and Stair-Edge Locomotion

***Abstract*— Wheels transport efficiently on level ground, whereas articulated legs place contacts across discontinuities; conventional wheel–leg hybrids commonly retain both mechanisms and their associated actuation. We study a different compromise: each of six 3-DoF legs terminates in one actuated, open C-shaped arc that is reused as a planted foot, a rolling contact, and a stair-edge hook. One controller runs the alternating-tripod walking reference and continuous arc rotation on the same six distal joints at the same instant: those joints are placed in velocity mode and the rolling rate is superposed on the differentiated walking reference, so neither a mode switch nor a hardware change is required. In a 500-Hz MuJoCo model, a same-platform ablation at a commanded $0.18\,\mathrm{m\,s^{-1}}$ yielded $0.117\,\mathrm{m\,s^{-1}}$ for articulated walking, $0.090\,\mathrm{m\,s^{-1}}$ for distal-only rotation, and $0.200\,\mathrm{m\,s^{-1}}$ for full-body rolling. A closed-wheel moment balance places a $122.5\,\mathrm{mm}$ riser limit on rotation over a rim; on three three-step geometries, rotation alone nevertheless cleared a $0.15\,\mathrm{m}$ riser, which isolates the open sector acting as a hook, whereas only the coordinated controller reached the top of the $0.20\,\mathrm{m}$ riser with $0.35\,\mathrm{m}$ tread, which isolates articulation. Walk–roll and roll–walk mode switches completed in 0.342 and $0.660\,\mathrm{s}$. These deterministic simulation trials establish a reproducible operating envelope and isolate the benefit of coordinating articulation with the shared arc contact. The study is a simulation-scope design analysis and does not yet validate physical speed, endurance, or stair reliability.**

I. INTRODUCTION

Level-ground speed and discontinuous-terrain mobility reward different contact mechanics. A wheel preserves a rolling constraint and avoids repeated foot placement; an articulated leg can place a contact and move the body relative to that contact. Hybrid robots often carry both a leg and a driven wheel [1], [2]. This increases capability, but the wheel remains a dedicated subsystem. At the opposite extreme, rotating-leg robots obtain impressive intrinsic mobility from few actuators [3]–[5], but their contact path and body motion are strongly coupled.

We ask a narrower mechanical question: *can the last actuated link of a fully articulated hexapod be shaped so that the same joint and contact surface support foot placement, rolling transport, and stair-edge engagement?* The resulting platform has 18 actuators—three per leg—and an open, C-shaped TPU arc as each distal link (Fig. 1). No wheel is deployed and no hardware transforms between modes. Instead, the contact interpretation changes: the lowest arc patch is held stationary during walking, the patch advances around the arc during rolling, and an arc boundary or tread patch reacts against a stair nosing during ascent.

This paper contributes: 1) a multifunctional arc–leg morphology with an explicit geometric model of its rolling and stair-edge contacts, including a closed-form riser bound that no amount of rotation can overcome; 2) SCONE-gait, a controller that runs the tripod walking reference and continuous arc rotation *simultaneously* on the same six distal joints by superposing a velocity-mode rolling rate on the differentiated walking reference, together with a phase-gated variant whose smoothness and duty-averaged transport fraction we characterize in closed form; and 3) a same-model benchmark with controlled restrictions —articulation only, distal rotation only, and the combined controller—in which the analytical contact-speed model predicts the measured slip of the rolling condition to within 5%.

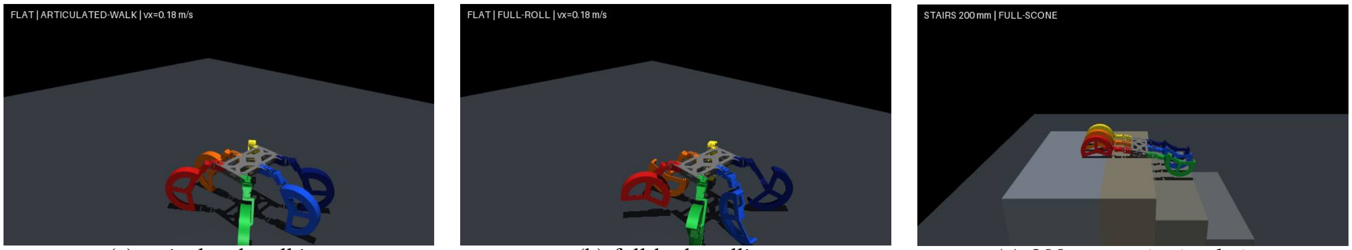
The scope of the evidence is stated once, here. All numbers in this paper come from a deterministic MuJoCo model; the nominal flat, stair, and transition conditions are single trials and the uneven-terrain condition uses three seeds. They are sufficient to separate the three control regimes on one model and to test the geometric bounds, and they are not sufficient for confidence intervals or for any physical performance claim. Physical demonstrations of the platform exist but are deliberately excluded from every numerical statement below.

II. RELATED WORK AND PROBLEM FORMULATION

A. Rotating Legs and Wheeled Legs

RHex showed that one continuously rotating compliant leg per hip can combine mechanical simplicity with rugged mobility under a clock-driven tripod gait [3], [6], and the X-RHex family carried that morphology into a research platform [7]. Whegs abstracted compliant, multi-spoke appendages and coupled propulsion to reduce actuator count [4]; Q-Whex later used quasi-wheels and reported stair climbing with geometry-dependent phase patterns [5]. The half-circle RHex leg is the closest prior morphology to ours and the distinction must be stated precisely: an RHex leg is a compliant C driven by a single hip rotation, so its contact point and the body pose are determined by the same input. Our C is the third joint of a 3-DoF leg, so the leg can hold the arc still, place it, or spin it while the body is posed independently, and it is this separation—not the C shape—that the ablation of Sec. VI tests.

A second family transforms the mechanism itself. Wheel Transformer deploys passive spokes when a wheel meets an obstacle [8], while Quattroped and TurboQuad change a leg into a wheel with a dedicated transformation degree of freedom [9], [10]. These recover both regimes at the cost of a transformation mechanism and its actuation. A third family keeps a wheel drive alongside a full leg: Cassino Hexapod III places omni-wheels at anthropomorphic leg tips [1],



(a) articulated walking (b) full-body rolling (c) 200-mm stair simulation

Fig. 1. One actuated C-shaped distal link is reused in three contact regimes. Images are reproducible MuJoCo captures from the benchmark conditions reported in Sec. VI; no physical-performance claim is inferred from the renderings.

ANYmal on Wheels integrates wheel constraints into whole-body optimization [2], [11], [12], and two-wheeled jumpers exploit the same separation in a smaller footprint [13]. They demonstrate the value of simultaneous wheel and leg control, but retain a wheel drive in addition to the leg chain. Our design does not claim that hybrid locomotion is new. The distinction tested here is *actuator and contact reuse*: the distal leg joint itself produces the foot trajectory, the rolling motion, and the edge engagement of one incomplete arc, with no transformation state and no additional drive.

Learned controllers have become the standard way to obtain robust legged locomotion [14]–[17], and residual formulations keep a structured reference while learning only a correction [18], [19]. We use a residual policy only in the low-speed regime of the deployment stack and, as Sec. IV-D makes explicit, it is inactive at the operating point of every experiment reported here.

B. Operating-Envelope Problem

Let the body command be

$$\vec{c} = [v_x,\ v_y,\ \omega_z]^\top \in \mathcal{C}, \qquad m \in \{W, R, C\}, \qquad (1)$$

for walking, rolling, and climbing. Let $\vec{q} \in \mathbb{R}^{18}$ be the actuated joint vector and Γ_i the active contact subset of distal arc i. The design seeks one morphology $\mathcal{M}$ and mode-dependent control laws π_m rather than a transformed mechanism $\mathcal{M}_m$:

$$\vec{q}_k^{\star} = \pi_m(\vec{x}_k, \vec{c}_k; \mathcal{M}), \quad \mathcal{M}_W = \mathcal{M}_R = \mathcal{M}_C. \qquad (2)$$

The relevant outcome is an operating envelope, not a single maximum speed,

$$\mathcal{O} = \{\bar{v}_x, e_v, C_{\text{mt}}, D_{\text{slip}}, P_{\text{top}}(h, d), t_{\text{top}}, t_{\text{switch}}\}, \qquad (3)$$

where h, d are riser and tread dimensions. We test three hypotheses. H1: coordinated rolling reaches a higher mean forward speed than either walking or distal-only rotation on the same model. H1 is deliberately stated as speed and not as command tracking, because a controller that overshoots a fixed command scores worse on e_v while travelling faster; both quantities are reported separately in Sec. VI. H2: coordinated articulation expands the climbable stair set beyond the geometric limit of arc rotation alone. H3: mode switching does not violate a 60° tilt bound.

We also define the qualitative column of Table I. Let J_i map leg i's joint rates to the pair (tangential advance of the contact patch along the distal surface, position of that contact relative to the body). Authority is *coupled* when rank $J_i = 1$, so that one input fixes both; *separated* when a dedicated drive makes the two independent at the cost of an extra actuator; and *partly separated* when, as here, three leg joints can choose the pair over a bounded set but the distal joint is shared between them.

TABLE I

MORPHOLOGICAL POSITION OF THE PROPOSED PLATFORM

Platform	leg actuation	rolling element	extra wheel drive	body/contact authority
RHex [3]	1/leg	compliant arc	no	coupled
Whegs I [4]	coupled	3-spoke wheg	shared	coupled
Q-Whex [5]	1/leg	quasi-wheel	no	coupled
Cassino III [1]	articulated	omni-wheel	yes	separated
Proposed	3/leg	open distal arc	no	partly separated

TABLE II

ACTUATION, CONTACT AND DISTAL GEOMETRY USED IN SIMULATION

Quantity	Value	Role
IDs 1–6	MX-28AT, 77 g	proximal
IDs 7–12	XM430-W350-T, 82 g	middle
IDs 13–18	XM430-W210-T, 82 g	distal arc
FR07 / FR12	20/46 g	frames
Arc radii r_i/r_o	112.5/122.5 mm	inner/outer
Arc width / opening ψ	44 mm / 135°	TPU contact
Sliding friction μ	1.0	arc–ground
Contact softness	3 mm, $\zeta = 1$	`solimp`/`solref`
Modeled mass	4.161 kg	MuJoCo only
Physics / control	500/50 Hz	integrate / target

III. PLATFORM AND CONTACT MODEL

A. Mechanical and Simulation Model

The six legs are numbered front to rear as right $1, 3, 5$ and left $2, 4, 6$. Actuator IDs 1–6 rotate the proximal links, IDs 7–12 the middle links, and IDs 13–18 the distal arcs. The MuJoCo model has 57 bodies, one floating base and 18 revolute joints, and a total modeled mass of 4.161 kg. Its component masses encode measured aluminum frames and actuators plus estimated 15%-infill polymer parts; therefore the total is a simulation parameter until the reassembled robot is weighed. Table II distinguishes that model quantity from verified part data.

Each actuator is a voltage-input DC motor with position feedback rather than an ideal torque source. For motor constants K, R, the outer PD loop and voltage saturation

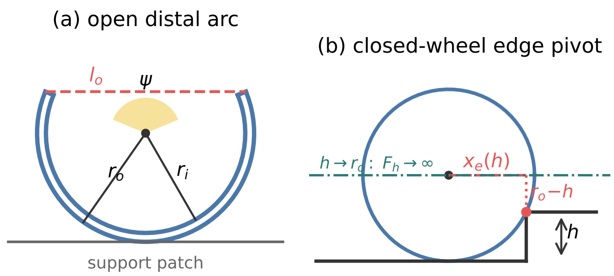


Fig. 2. Contact geometry of one distal link. (a) The open sector: outer and inner radii r_o, r_i, opening angle ψ and opening chord l_o, with the lowest patch acting as the support point during walking. (b) The closed-wheel pivot used as a reference line: the moment arm is $x_e(h)$ and the balancing horizontal force diverges as $h \to r_o$, so any measured ascent above h_{cw} must come from the opening or from articulation.

are

$$\tau_i^c = k_{p,i}(q_i^\star - q_i) + k_{d,i}(\dot{q}_i^\star - \dot{q}_i), \tag{4}$$

$$V_i = \text{sat}_{[-V_n, V_n]}\left(\frac{R_i}{K_i}\tau_i^c + K_i\dot{q}_i\right). \tag{5}$$

Each group is specified by its rated triple (voltage, stall torque, no-load speed), from which $K = V_n/\dot{q}_{\text{nl}}$ and $R = KV_n/\tau_{\text{stall}}$ follow: $(12\,\text{V}, 2.5\,\text{N m}, 5.76\,\text{rad/s})$, $(12, 4.1, 4.82)$, and $(12, 3.0, 8.06)$. The simulated gains (k_p, k_d) in SI units are $(5.73, 0.752)$, $(9.40, 0.792)$, and $(6.88, 0.264)$. Integral gain is zero; supply voltage and a stall-matched saturation torque bound the response.

B. Arc Contact and Stair-Edge Mechanics

Figure 2 fixes the notation. For outer radius r_o, inner radius r_i, and opening ψ, the opening chord is

$$l_o = 2r_o \sin(\psi/2) = 0.226 \text{ m}. \tag{6}$$

Contacts use a six-dimensional friction cone with sliding coefficient $\mu = 1.0$ and a 3 mm compliant layer, so the TPU arc is modeled as soft but not deformable. During walking, the support point is the centroid of mesh vertices within 0.1 mm of the lowest arc patch. During rolling, this patch advances with the distal angle.

Consider a single arc pivoting over a sharp riser of rise h with its centre initially one radius above the lower tread. The horizontal centre-to-edge offset is

$$x_e(h) = \sqrt{r_o^2 - (r_o - h)^2} = \sqrt{2r_o h - h^2}, \quad 0 \le h \le 2r_o. \tag{7}$$

If that arc carries load F_i, safety factor $\gamma \ge 1$ and drive efficiency $\eta \le 1$ give a quasi-static lower bound on the pivot torque, while balancing the same gravity moment with a horizontal force at the arc centre requires

$$\tau_{e,i} \ge \gamma\,\frac{F_i\,x_e(h)}{\eta}, \tag{8}$$

$$F_{h,i} \ge \gamma\,\frac{F_i\,x_e(h)}{r_o - h}, \qquad \mu_{\min} = \frac{|F_t|}{F_n}. \tag{9}$$

Equation (9) is the operative bound for a *closed* wheel of the same radius: $F_{h,i} \to \infty$ as $h \to r_o^-$, and the moment balance has no solution for $h \ge r_o$. With a nosing radius r_n and clearance c the closed-wheel riser limit is therefore

$$h < h_{\text{cw}} = r_o - r_n - c \le 122.5 \text{ mm}. \tag{10}$$

Both the 150 mm ($1.22r_o$) and the 200 mm ($1.63r_o$) risers of Sec. VI lie above h_{cw}, so any ascent of either one is produced by a mechanism the closed wheel does not have. Two such mechanisms exist on this platform, and the ablation is designed to tell them apart. The first is the opening itself: a riser whose height is smaller than the opening chord, $h < l_o = 226$ mm, can enter the 135° gap and be engaged by the far end of the C rather than pivoted over by the rim, so the arc acts as a hook. The second is articulation, which raises and repositions the whole arc irrespective of edge geometry. Equation (10) is thus not a prediction of failure but a *reference line*: it converts each measured ascent above 122.5 mm into evidence about which of the two mechanisms is responsible. A second dimensionless descriptor compares the opening chord with one step's diagonal, $l_o/\sqrt{h^2 + d^2}$, giving 0.87, 0.91, and 0.56 for the three tested profiles. We report it because it is the standard legged-wheel comparison, and note that it does not by itself order the observed outcomes.

IV. SCONE-GAIT: SIMULTANEOUS WALKING AND ARC ROTATION

The design goal is a single controller in which the tripod walking motion and the continuous rotation of the distal arcs run *at the same time* on the same six joints, rather than a mode switch between a walking machine and a driving machine. Section IV-A defines the bounded reference shared by every condition, Sec. IV-B defines the controller that produced the results in Sec. VI, and Sec. IV-C states a phase-gated variant of the same idea together with the properties it guarantees.

A. Alternating-Tripod Reference

Tripods $\mathcal{T}_A = \{1, 4, 5\}$ and $\mathcal{T}_B = \{2, 3, 6\}$ have offsets $\delta_i \in \{0, 1/2\}$. With cycle frequency f and duty factor D, the leg phase is

$$\phi_i(t) = (ft + \delta_i) \bmod 1. \tag{11}$$

At nominal foot position $\vec{p}_{i0} = [x_i, y_i, z_i]^\top$, the planar velocity induced by the commanded body twist and the resulting stance stroke are

$$\vec{u}_i = \begin{bmatrix} v_x - \omega_z y_i \\ v_y + \omega_z x_i \end{bmatrix}, \quad \vec{s}_i = D\vec{u}_i/f. \tag{12}$$

The elliptical workspace projection $\vec{s}_i \leftarrow \vec{s}_i/\max(1, \|[s_{ix}/s_x^{\max}, s_{iy}/s_y^{\max}]\|_2)$ prevents unreachable strides. Let $S(a) = 10a^3 - 15a^4 + 6a^5$ and $H(a) = 16a^2(1-a)^2$. The desired contact trajectory is

$$\vec{p}_i^\star = \vec{p}_{i0} + \begin{cases} (\frac{1}{2} - \phi_i/D)[\vec{s}_i^\top, 0]^\top, & \phi_i < D, \\ (S(a) - \frac{1}{2})[\vec{s}_i^\top, 0]^\top + h_f H(a)\vec{e}_z, & \phi_i \ge D, \end{cases} \tag{13}$$

where $a = (\phi_i - D)/(1 - D)$. Damped numerical inverse kinematics maps the six targets to the bounded 18-joint reference $\vec{q}^{\text{ref}}$. Two parameter sets of (11)–(13) are used and are reported here because they are not identical: the walking condition uses $(f, D, h_f, s_x^{\max}, s_y^{\max}) = (1.0\,\text{Hz}, 0.50, 25, 90, 70\,\text{mm})$ and the rolling conditions use $(0.8\,\text{Hz}, 0.58, 20, 55, 45\,\text{mm})$, the smaller stroke reflecting that rotation, not stride, is expected to carry most of the displacement.

B. Superposed Rotation on Velocity-Mode Distal Joints

Joints 1–12 track $\vec{q}^{\text{ref}}$ in position mode. The six distal joints are switched to velocity mode, which removes the one-turn bound, and are commanded as the sum of a continuous rolling term and the differentiated walking reference,

$$\dot{q}_{d,i} = \underbrace{\dot{\theta}_i^{\text{roll}}}_{\text{transport}} + \lambda \underbrace{\Delta q_{d,i}^{\text{ref}}/\Delta t}_{\text{walking}}, \qquad \lambda = 0.35, \tag{14}$$

which is the concrete form of "walk and rotate at once": neither term is gated off by the other. Let the normalized command activity be

$$\alpha(\vec{c}) = \text{clip}\left(\max_j \frac{|c_j|}{c_j^{\max}}, 0, 1\right), \quad \vec{c}^{\max} = [0.18, 0.12, 0.9], \tag{15}$$

and let the stance/swing rate ratio be the two-level function

$$\kappa_i(\phi_i) = \begin{cases} \kappa_{\text{st}} = 0.80, & i \in \text{stance}, \\ 1, & i \in \text{swing}. \end{cases} \tag{16}$$

With polarity $\sigma_i \in \{-1, 1\}$ and alignment $A_i \in [0, 1]$ obtained from a finite-difference contact tangent $\vec{t}_i = \partial \vec{p}_{c,i}/\partial q_{d,i}$ measured on the mesh,

$$A_i = \max\left(0,\ \hat{\vec{t}}_i^{\top} \frac{-\vec{u}_i}{\|\vec{u}_i\|}\right), \tag{17}$$

the unfiltered rolling rate and its first-order realization are

$$\omega_i^{\text{cmd}} = -\sigma_i\, \omega_0\, \alpha(\vec{c})\, A_i\, \kappa_i, \quad \omega_0 = 4.20\,\text{rad/s}, \tag{18}$$

$$\dot{\theta}_i^{\text{roll}} \leftarrow \dot{\theta}_i^{\text{roll}} + \left(1 - e^{-\Delta t/\tau_r}\right)\left(\omega_i^{\text{cmd}} - \dot{\theta}_i^{\text{roll}}\right), \tag{19}$$

with $\tau_r = 0.10\,\text{s}$. The walking term of (14) is clipped to $\pm 110°/\text{s}$ and low-pass filtered with $\tau_b = 0.04\,\text{s}$ so that an inverse kinematics branch change cannot appear as a rate spike. At start-up the three tripod-B arcs are staggered by $60°$; without this offset all six openings can face the ground simultaneously and the body drops.

The consequence that matters for Sec. VI is that (18) is an *activity-scaled* rate, not a no-slip rate. The commanded contact speed during stance is

$$v_c = r_o\, \omega_0\, \alpha\, \kappa_{\text{st}} = 0.411\ \text{m}\,\text{s}^{-1} \quad \text{at } \alpha = 1, \tag{20}$$

so whenever the body advances more slowly than v_c the difference must appear as measured contact slip,

$$D_{\text{slip}} \approx \int_0^T \left| v_c - \|\vec{v}_{xy}(t)\| \right| dt. \tag{21}$$

Section VI uses (21) as a check rather than as a claim.

C. Phase-Gated Multi-Turn Variant

Replacing the two-level ratio κ_i of (16) by a smooth gate lets the arc be held still while it first accepts load. For stance progress $p = \phi_i/D$, swing progress a, point-support ratio $\rho = 0.55$ and swing hold ratio $\nu = 0.70$, define

$$g_i = \begin{cases} 0, & p \le \rho, \\ S\left(\frac{p-\rho}{1-\rho}\right), & \rho < p \le 1, \\ 1, & a \le \nu, \\ 1 - S\left(\frac{a-\nu}{1-\nu}\right), & a > \nu, \end{cases} \tag{22}$$

and drive the arc at the no-slip rate rather than a fixed rate,

$$\dot{\Theta}_i^{\text{ideal}} = \min\left(\frac{\|\vec{u}_i\|_2}{r_o}, \dot{\Theta}_{\max}\right), \tag{23}$$

$$\Theta_{i,k+1} = \Theta_{i,k} - \sigma_i A_i g_i \dot{\Theta}_i^{\text{ideal}} \Delta t, \quad q_{d,i}^{\star} = q_{\text{IK},d,i}^{\star} + \Theta_i. \tag{24}$$

Because $S(0) = S'(0) = S''(0) = 0$ and $S(1) = 1$, $S'(1) = S''(1) = 0$, the four branches of (22) join with matching value, slope and curvature, so $g_i \in C^2$ and the commanded arc acceleration is continuous at every gate boundary. Two further properties follow directly from the parameters. First, each arc is stationary for the first 55% of its stance, so the patch that accepts load does not move while it does so. Second, with the deployment duty factor $D = 0.60$ and offsets $\{0, \frac{1}{2}\}$ the propulsive windows of the two tripods are $(\rho D, D) = (0.33, 0.60)$ and $(0.83, 1.10) \bmod 1$, which are disjoint: the two tripods never rotate under load simultaneously. Averaging (22) over one cycle gives the closed form

$$\bar{g} = \frac{D(1-\rho)}{2} + \frac{(1-D)(1+\nu)}{2} \overset{D=0.60}{=} 0.475, \tag{25}$$

so a gated arc delivers only the fraction $\bar{g}$ of the commanded displacement and the remainder must come from articulation. This variant is implemented and exercised by unit tests, but the benchmark of Sec. VI uses (14)–(19); (22)–(25) are stated here as the analysed design and are not evidence for any reported number.

D. Residual Policy and Speed Blend (Deployment Stack)

For completeness we state the low-speed supervisor used by the interactive deployment stack. Residual learning augments a structured reference rather than relearning the contact schedule [18], [19]. The 70-D observation contains body linear/angular velocity, projected gravity, 18 joint positions, 18 joint velocities, the previous 18-D action, the command, and phase and heading-error sine/cosine pairs; PPO optimizes the clipped surrogate [20]

$$L^{\text{clip}} = \mathbb{E}_t\left[\min(r_t \hat{A}_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\hat{A}_t)\right]. \tag{26}$$

At replay a smoothstep in translation speed $v = \|\vec{v}_{xy}\|$ hands authority from the policy to the analytical reference,

$$u = \text{clip}\left(\frac{v - v_0}{v_1 - v_0}, 0, 1\right), \quad \beta(v) = 3u^2 - 2u^3, \tag{27}$$

$$\begin{aligned} \vec{q}^{\star} &= (1-\beta)\vec{q}_{\text{ref}} + \beta \vec{q}_{\text{arc}} \\ &\quad + (1-\beta)\, \vec{s}_a \odot \vec{a}_{\text{PPO}}, \end{aligned} \tag{28}$$

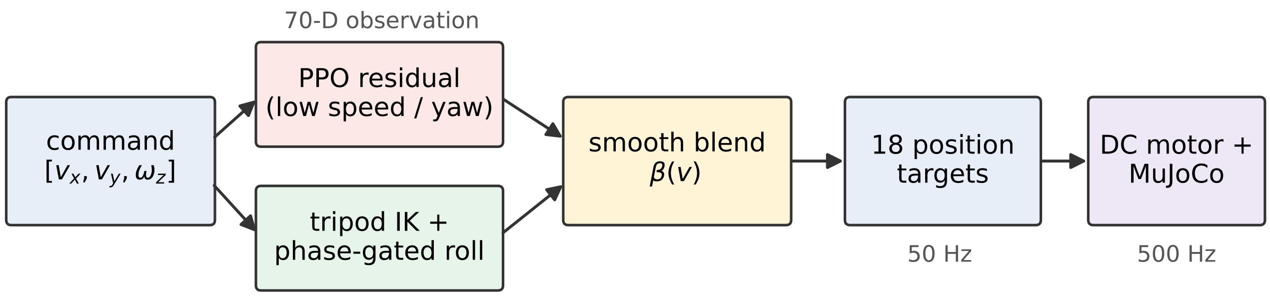


Fig. 3. Control paths. Joints 1–12 track the bounded tripod reference in position mode while joints 13–18 run in velocity mode and add continuous arc rotation to the differentiated reference, so walking and rotation are commanded at the same instant. The residual policy and its speed blend belong to the deployment stack and are inactive in the reported benchmark.

with $(v_0, v_1) = (0.10, 0.18)$ m/s. We state explicitly that $\beta(v_1) = 1$: at the $0.18\,\mathrm{m\,s^{-1}}$ command used throughout Sec. VI the residual term vanishes, and the benchmark controllers do not instantiate the policy at all. The residual is therefore not offered as evidence in this paper; it is reported because it defines the low-speed regime that the analytical controller does not cover.

E. Stair State Sequence

The stair controller executes Walk→Drive→Climb, verifies each settled pose, braces the front middle joints, and then synchronizes the six distal phases. A geometry-conditioned brace uses 180°, 184°, or 195° over increasing riser bands. This is a deterministic supervisor that is given the stair dimensions; it performs no terrain perception and no foothold search.

V. EXPERIMENTAL METHOD

A. Reproducible Simulation Protocol

Experiments use MuJoCo [21] with 2 ms integration and 20 ms control. Every trial creates a fresh floating-base model and controller. The nominal flat test settles for 1 s and measures 6 s at $\vec{c} = [0.18, 0, 0]$. The three same-model restrictions are:

- A articulated tripod walking, distal joints in position mode with the continuous rolling term of (14) removed;
- B distal-only rotation, distal joints in velocity mode driven by (18) while the body and proximal targets are held at the staggered stance pose;
- C the full controller of Sec. IV-B, in which both terms of (14) are active simultaneously.

The restrictions therefore isolate control authority on one unchanged model. Two caveats follow from Sec. IV-A and are stated before the results rather than after them: A and C do not share the walking parameter set (1.0 Hz, $D = 0.50$, 90 mm stroke versus 0.8 Hz, $D = 0.58$, 55 mm stroke), and the rolling rate of (18) is scaled by normalized command activity rather than tuned per condition. Neither condition was separately optimized for maximum speed, so the comparison establishes the qualitative ordering of the three regimes on one model and not the best speed attainable by any of them.

The stair set contains three 3-step profiles: $(h, d) = (0.10, 0.24)$, $(0.15, 0.20)$, and $(0.20, 0.35)$ m, where d is the minimum tread. Strategies are distal-only rotation, which holds every non-arc joint and spins the six arcs at one constant rate so that no articulation is available; a synchronized open-loop roller with a fixed front brace; and the full geometry-conditioned controller. Reaching the upper landing defines success. Transition trials measure the time from the first mode-change command until all guarded actuator conditions are satisfied, followed by 2 s recovery. Uneven-terrain smoke tests use three randomized seeds and 2 s windows; they check execution, not publication-grade robustness.

B. Metrics

Body-frame velocity tracking error and integrated slip are

$$e_v = \sqrt{\frac{1}{T}\int_0^T \|\vec{v}_{xy}(t) - \vec{c}_{xy}\|_2^2 dt}, \tag{29}$$

$$D_{\text{slip}} = \int_0^T \frac{1}{|\mathcal{K}(t)|} \sum_{k\in\mathcal{K}(t)} \|\vec{v}^{\parallel}_{c,k}(t)\|_2 dt, \tag{30}$$

where $\mathcal{K}$ contains loaded arc–ground contacts carrying more than 1 N, about 2.5% of the modeled body weight. Because e_v is measured against a constant command, a controller that overshoots that command is penalized; $\bar{v}_x$ and e_v are therefore read together. Absolute mechanical work and mechanical cost of transport are

$$W_{\text{abs}} = \int_0^T \sum_{j=1}^{18} |\tau_j \dot{q}_j| dt, \qquad C_{\text{mt}} = \frac{W_{\text{abs}}}{mgd_{xy}}, \tag{31}$$

with d_{xy} the net planar displacement. An electrical estimate uses $I_j = (V_j - K_j\dot{q}_j)/R_j$ and $E_e = \int \sum_j |V_j I_j| dt$. Because neither battery conversion loss nor servo electronics is modeled, E_e is diagnostic, not physical battery energy. Uprightness is $u_z = \vec{e}_z^{\top} R_{WB} \vec{e}_z$; termination occurs below $\cos 60°$ or on non-finite state.

C. Scope of the Evidence

Every trial writes one record carrying the source revision, the seed, and all model scale factors, so each reported number is traceable to the configuration that produced it. The nominal

TABLE III

FLAT NOMINAL TRIAL WITH A 0.18 M/S FORWARD COMMAND

Controller	$\bar{v}_x$ m/s	e_v m/s	C_{mt} –	slip m
Articulated walk	0.117	**0.097**	**0.661**	**0.212**
Distal only	0.090	0.268	2.576	1.603
Full rolling	**0.200**	0.253	1.227	1.323

TABLE IV

STAIR TOP-REACHING OUTCOMES (ONE DETERMINISTIC TRIAL)

	100-mm rise		150-mm rise		200-mm rise	
Controller	top	s	top	s	top	s
Distal only	yes	4.886	yes	14.960	no	–
Open loop	yes	3.818	yes	5.546	no	–
Full controller	yes	2.646	yes	6.736	**yes**	5.964

flat, stair, and transition conditions are single deterministic trials and the uneven-terrain condition uses three seeds; they support reproduction and ablation ordering, not confidence intervals. No physical measurement enters any table, figure, or conclusion in this paper.

VI. RESULTS

A. Flat-Ground Ablation

Table III and Fig. 4 show different trade-offs. Coordinated rolling reached $0.200\,\mathrm{m\,s^{-1}}$, exceeding the $0.18\,\mathrm{m\,s^{-1}}$ command by 11.2% and travelling 71% faster than articulated walking, which supports H1 as stated. The same table refutes the stronger reading of H1 that we deliberately avoided: measured against the constant command, walking has the smallest tracking error ($e_v = 0.097$ against 0.253 and $0.268\,\mathrm{m\,s^{-1}}$), because rolling overshoots the command rather than following it. Rolling also consumed $3.18\times$ the absolute mechanical work and accumulated $6.23\times$ the contact slip of walking. Distal-only rotation was both slower and more expensive than walking, so the arc by itself does not explain the speed: in this model the gain comes from coordinating posture with rotation, and it is bought with slip and work rather than with efficiency.

The slip is not an unexplained artifact. Equation (20) fixes the commanded stance contact speed at $v_c = 0.411\,\mathrm{m\,s^{-1}}$ for $\alpha = 1$, which is about twice the body speed the rolling condition achieves, and (21) then predicts $|v_c - \bar{v}_x|T = 1.27\,\mathrm{m}$ of integrated slip over the six-second window against $1.323\,\mathrm{m}$ measured, an error of 4.2%. For distal-only rotation the same expression predicts $1.93\,\mathrm{m}$ against $1.603\,\mathrm{m}$ measured; it over-predicts by 20% because the alignment factor A_i of (17) falls below unity when the body does not follow the contact. The agreement identifies the cause of the slip precisely: the rolling rate of (18) is scaled by normalized command activity and not by the no-slip rate of (23), so a fixed fraction of the commanded surface speed is dissipated at the contact. This is a property of the controller that produced these numbers, and it is the first quantity the phase-gated variant of Sec. IV-C is intended to remove.

B. Stair Coordination Ablation

Table IV separates the three climbing mechanisms of Sec. VI along the reference line of (10). At $100\,\mathrm{mm}$, below the closed-wheel limit $h_{\mathrm{cw}} = 122.5\,\mathrm{mm}$, every strategy succeeded, as a wheel of the same radius would. At $150\,\mathrm{mm}$ ($1.22 r_o$), which no closed wheel of this radius can pivot over, distal-only rotation still reached the landing in $14.960\,\mathrm{s}$ with no articulation whatsoever; since the riser is well inside the opening chord ($150 < l_o = 226\,\mathrm{mm}$), this isolates the open sector acting as an edge hook and is the clearest evidence in this study that the opening, not merely the distal degree of freedom, carries function. At $200\,\mathrm{mm}$ ($1.63 r_o$) both restricted strategies failed and only the coordinated controller reached the top, in $5.964\,\mathrm{s}$ with $88.18\,\mathrm{J}$ of work, which places the remaining capability on articulation. Speed and safety did not coincide: at $150\,\mathrm{mm}$ the open-loop condition was fastest ($5.546\,\mathrm{s}$) but fell to $u_z = 0.459$, close to the $\cos 60°$ termination boundary, whereas the coordinated controller held $u_z = 0.755$.

C. Transitions and Uneven-Terrain Smoke Test

Walk-to-roll and roll-to-walk guards completed in 0.342 and $0.660\,\mathrm{s}$. Minimum uprightness over the full transition/recovery windows was 0.991 in both directions, and neither trial produced forbidden collisions or IK failures, so H3 was not violated in either direction. On randomized uneven terrain, all nine short trials executed without the tilt termination: mean $\bar{v}_x$ over three seeds was 0.0768 ± 0.0043, 0.1025 ± 0.0356, and $0.2009 \pm 0.0126\,\mathrm{m\,s^{-1}}$ for walk, distal-only, and full rolling. The sample size and $2\,\mathrm{s}$ window do not establish robust-terrain superiority.

VII. DISCUSSION

The experiments answer a more precise question than "can a hybrid robot walk and drive." Using the closed-wheel bound (10) as a reference line, the three stair outcomes decompose the climbing capability of one distal link into three contributions: a rim pivot below $122.5\,\mathrm{mm}$ that any wheel of the same radius would also provide; an edge hook that carried a $150\,\mathrm{mm}$ riser with no articulation at all, which is attributable to the $135°$ opening; and body elevation by the other two leg joints, without which the $200\,\mathrm{mm}$ riser was not climbed by any restriction. On flat ground the picture is the mirror image: rotation alone was slower and more expensive than walking, and the speed gain appeared only when rotation and posture were commanded together. Taken together these support contact reuse as the contribution—a foot, a quasi-wheel and an edge hook are three control regimes of one link rather than three installed mechanisms—and they locate where each regime stops being sufficient.

The costs are equally explicit. Rolling produced $6.23\times$ the slip and $3.18\times$ the mechanical work of walking, and (20) shows why: the executed controller commands an activity-scaled surface speed rather than the no-slip speed, so roughly half of the commanded surface motion is dissipated at the

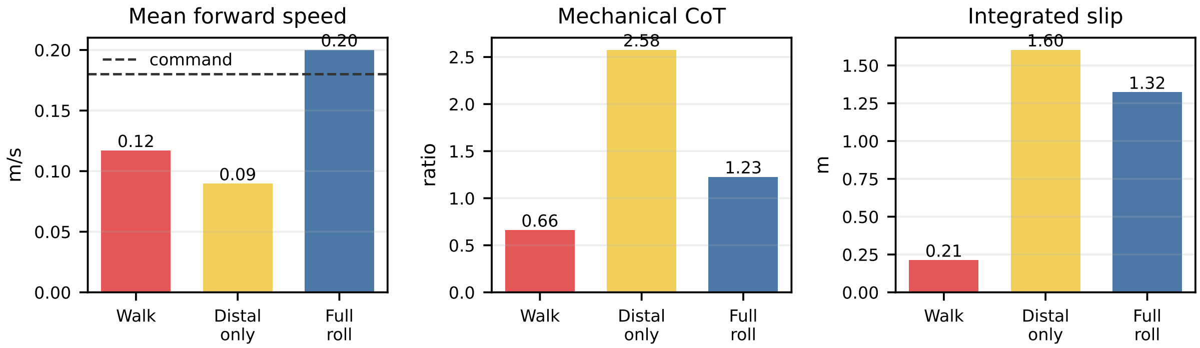


Fig. 4. Same-model flat-ground development trial (one run per controller, six-second window). Values are descriptive; error bars are intentionally absent.

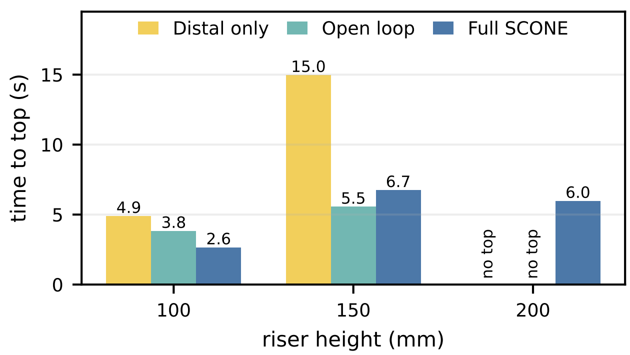


Fig. 5. Time to the upper landing for three controller restrictions over three 3-step geometries (one deterministic run each). Bars marked *no top* did not reach the landing; the binary outcomes are tabulated in Table IV.

contact. The present model therefore supports a speed trade-off, not an energy-efficiency claim. The phase-gated variant of Sec. IV-C targets exactly this term, and its duty-averaged transport fraction (25) predicts that it trades some of the speed back for the slip it removes; testing that prediction is the immediate next experiment.

Four limitations bound how far these results generalize. First, the evidence is a deterministic simulation: TPU compliance and friction, polymer infill mass, gearbox efficiency, backlash, thermal limits and battery sag are approximated, and the modeled mass remains a parameter until the assembled robot is weighed. Second, the nominal conditions are single trials, so the reported differences order the three regimes on one model but carry no confidence intervals. Third, the restrictions are not tuned against each other: the walking and rolling conditions use different cycle frequencies and stroke limits (Sec. IV-A), so the comparison is between regimes rather than between best-tuned controllers. Fourth, the hook interpretation of the $150\,\mathrm{mm}$ result is inferred from geometry and from the absence of articulation in that condition; a same-radius closed-wheel control, which the present model does not include, would settle it directly. The stair supervisor is also given its geometry and performs no perception or foothold search.

VIII. Conclusion

We presented an 18-actuator hexapod whose six open arc-shaped distal links are reused for planted support, rolling contact and stair-edge engagement, together with a controller that commands tripod walking and continuous arc rotation on the same joints at the same instant. In deterministic MuJoCo trials the coordinated controller reached $0.200\,\mathrm{m\,s^{-1}}$ where articulated walking reached $0.117\,\mathrm{m\,s^{-1}}$; its analytical contact-speed model predicted the measured slip to within 4.2%; and the three stair outcomes separated rim pivoting, edge hooking and articulated elevation against a closed-wheel bound of $122.5\,\mathrm{mm}$. Guarded mode switches stayed upright and completed in under $0.7\,\mathrm{s}$. These results establish a simulation operating envelope and a falsifiable physical test plan. They do not yet validate physical speed, endurance, energy efficiency or stair reliability.

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